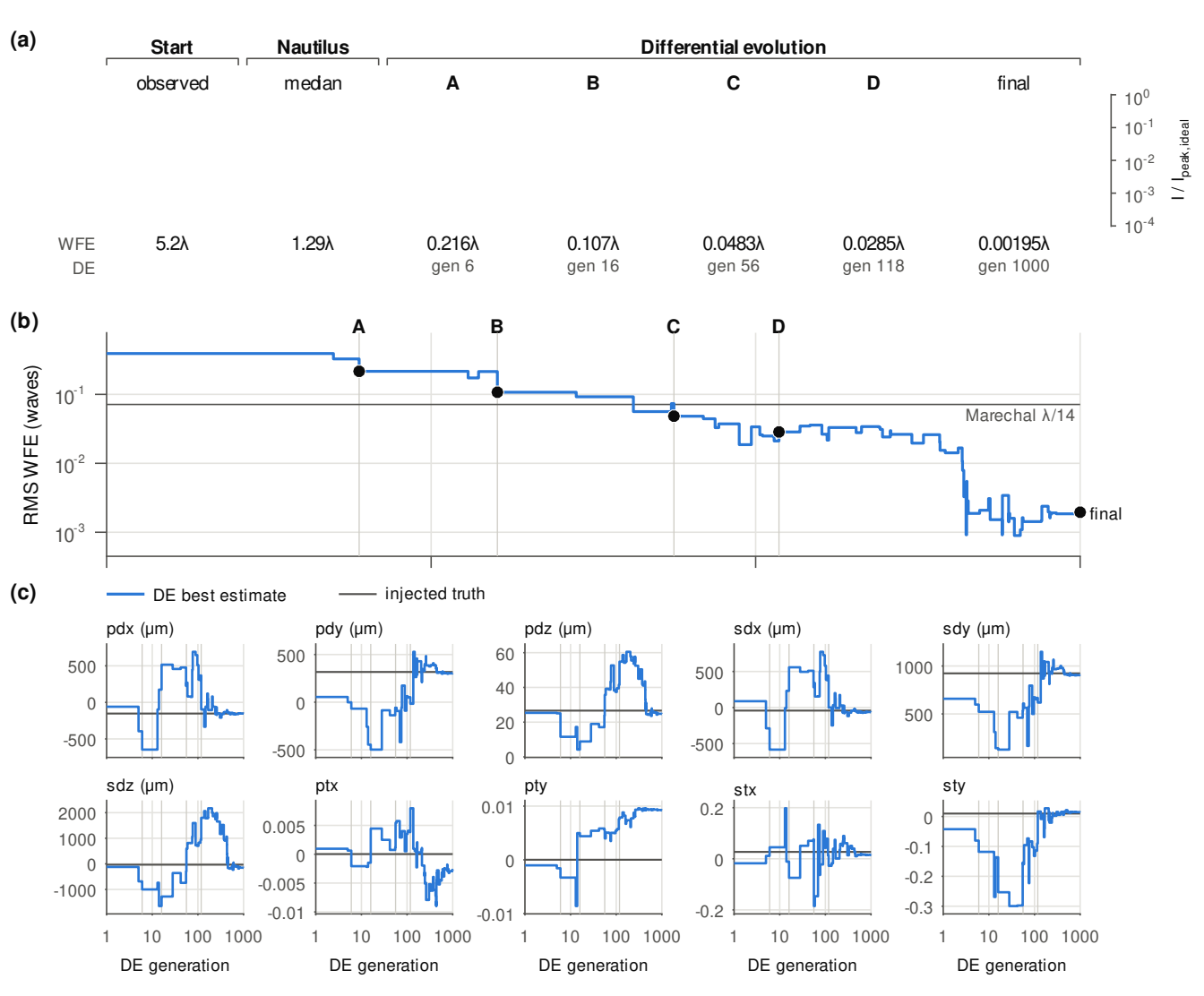

(a)
Start
Nautilus
Differential evolution
observed
medan
A
B
C
D
final
WFE
DE
5.2λ
1.29λ
0.216λ
gen 6
0.107λ
gen 16
0.0483λ
gen 56
0.0285λ
gen 118
0.00195λ
gen 1000
I / I peak,ideal
(b)
RMS WFE (waves)
Marechal λ/14
final
(c)
DE best estimate
injected truth
pdx (μm)
pdy (μm)
pdz (μm)
sdx (μm)
sdy (μm)
sdz (μm)
ptx
pty
stx
sty
DE generation